`bispectrum`: Selective G-Bispectra Made Practical

Anonymous Author(s)
Affiliation
Address
email

Abstract

TODO: Write a clean 120–150 word abstract after we decide the final empirical scope.

We present `bispectrum`, a PyTorch library for computing selective G-bispectral invariants across seven group/domain pairs as differentiable `nn.Modules`. Building the library required two technical contributions: (i) a selective $\mathrm{SO}(3)$ bispectrum on S^2 that achieves $O(L^2)$ coefficients—matching the information-theoretic lower bound—instead of $O(L^3)$ for the full bispectrum, and (ii) resolving a previously undocumented inversion failure for the octahedral group via bootstrap initialization followed by Levenberg–Marquardt refinement. We evaluate the library across three group/domain settings—C_8 on PCam, octahedral on OrganMNIST3D, and $\mathrm{SO}(3)$ on Spherical MNIST—demonstrating exact rotation invariance and the data-efficiency advantage of complete bispectral invariants over incomplete alternatives such as power spectrum and norm pooling.

1 Introduction

Equivariant neural networks [Cohen and Welling, 2016, Weiler and Cesa, 2019] enforce symmetry by construction, but they still require an invariant representation at the point where a prediction must no longer depend on orientation. In practice, invariant pooling remains a weak point: standard choices such as norm pooling, gate-based pooling, or max/sum aggregation are usable [Weiler and Cesa, 2019], but they do not preserve all equivariant information. The bispectrum offers a mathematically principled alternative, since for many groups it yields exact invariants with reconstruction guarantees [Kakarala, 1992].

The obstacle is practical rather than conceptual. Existing bispectral constructions are scattered across theory papers, domain-specific derivations, and small implementations, with no general-purpose PyTorch library covering the finite-group and compact-group cases relevant to modern geometric deep learning. Even for groups where selectivity reduces the coefficient count from $O(|G|^2)$ to $O(|G|)$ [Mataigne et al., 2024], a user still needs a numerically stable implementation, a consistent API, inversion routines, tests, and benchmarked runtime behavior.

This paper presents `bispectrum`, a PyTorch library for computing bispectral invariants and, where available, their inverses. The library supports cyclic, dihedral, torus, disk, sphere, and octahedral settings as differentiable `nn.Modules` intended to plug into standard ML workflows. Beyond the software contribution, the paper makes two technical contributions that arose during implementation: a selective $\mathrm{SO}(3)$ bispectrum on S^2 with $O(L^2)$ coefficients (Section 3.4), and a resolution of the bootstrap inversion failure [Mataigne et al., 2024] for the octahedral group (Section 3.5).

TODO: Rewrite the introduction once the final empirical scope is fixed.

Contributions.

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Table 1: Invariant nonlinearities for equivariant CNNs. The selective bispectrum is the only method that is simultaneously exact, complete, and linear-cost for finite groups.

Nonlinearity	Exact?	Complete?	Coefficients	Key limitation
Norm	Yes	No	$O(\|G\|)$	Discards phase
Gate	Yes	No	$O(2\|G\|)$	Scalar bottleneck
Fourier pointwise	No	Yes	$O(\|G\| \log \|G\|)$	Aliasing
Full bispectrum	Yes	Yes	$O(\|G\|^2)$	Quadratic cost
Sel. bispectrum	**Yes**	**Yes**	$\mathbf{O(\|G\|)}$	**(this work)**

1. **Software contribution.** A PyTorch library implementing bispectral invariants for multiple group/domain pairs with a consistent `nn.Module`-style interface.
2. **Selective $\mathrm{SO}(3)$ bispectrum on S^2.** We construct and implement a selective bispectrum for $\mathrm{SO}(3)$ acting on S^2 that uses $O(L^2)$ coefficients instead of the full $O(L^3)$, matching the information-theoretic lower bound of $(L+1)^2 - 3$ independent invariants. This is, to our knowledge, the first $O(L^2)$ complete invariant for this setting.
3. **Octahedral inversion.** An analysis of why the standard selective-bispectrum bootstrap inversion fails for the octahedral group, together with a working bootstrap + Levenberg–Marquardt correction.
4. **Empirical validation.** A task-driven evaluation of the library as an invariant pooling primitive across three group/domain settings, demonstrating the data-efficiency advantage of complete bispectral invariants.

2 Mathematical Background

2.1 Equivariant features and invariant maps

A neural network layer is *G-equivariant* if its output transforms predictably under the group action: $[\pi_{\mathrm{out}}(g) \circ \Phi](f) = \Phi(\pi_{\mathrm{in}}(g)f)$ for all $g \in G$ [Cohen and Welling, 2016, Weiler and Cesa, 2019]. Stacking equivariant layers preserves this property, but any task that requires a prediction independent of orientation—classification, regression, detection—must eventually map equivariant features to *invariant* ones.

Table 1 summarises the trade-offs. Norm pooling and gated nonlinearities are exact invariants but *incomplete*: they discard the phase of Fourier blocks, collapsing distinct orbits. Fourier-space pointwise nonlinearities preserve completeness but sacrifice exactness through aliasing [Weiler and Cesa, 2019]. The full bispectrum is exact and complete but costs $O(|G|^2)$ coefficients.

2.2 The G-bispectrum and selective bispectra

Let G be a compact group with irreducible representations $\{\rho_i\}_{i \in \hat{G}}$ of dimensions d_i, and let $\hat{f}(\rho_i) = \sum_{g \in G} f(g)\, \rho_i(g)$ denote the group Fourier transform. Given the Clebsch–Gordan decomposition $\rho_i \otimes \rho_j \cong \bigoplus_k m_{ij}^k \rho_k$ with intertwining matrix C_{ij}, the *bispectrum* is

$$\beta_{\rho_i,\rho_j} = C_{ij}^\top \left(\hat{f}(\rho_i) \otimes \hat{f}(\rho_j)\right) C_{ij}\, \hat{f}(\rho_k)^\dagger. \tag{1}$$

Each β_{ρ_i,ρ_j} is invariant under $f \mapsto f \circ g$ and, taken over all pairs (i, j), the bispectrum is a *complete* invariant: it separates orbits for generic signals [Kakarala, 1992]. However, the full set has $O(|G|^2)$ coefficients.

Selective bispectra. Mataigne et al. [2024] show that for finite groups, a BFS traversal of the Kronecker product table of $\hat{G}$ selects $O(|G|)$ bispectral coefficients that remain complete. This reduction from quadratic to linear is the key to practical deployment: for example, the octahedral group drops from 576 to 172 coefficients.

3 The `bispectrum` Library

3.1 Design principles

The library is organised around six principles: **(P1)** *Naming encodes the math*: class names follow `{Group}on{Domain}` (e.g. `SO3onS2`), making the mathematical context immediately explicit. **(P2)** *`nn.Module` only*: every computation is a `torch.nn.Module` with CG matrices, DFT kernels, and Bessel roots registered as non-learnable buffers so that `.to(device)` and `.to(dtype)` work transparently. **(P3)** *Selective by default*: all modules default to $O(|G|)$ selective coefficients; the full $O(|G|^2)$ set is available via `selective=False`. **(P4)** *Raw signals in*: modules accept real-valued inputs in the natural domain and handle the Fourier transform internally. **(P5)** *Inversion on-module*: `bsp.invert(beta)` returns the reconstructed signal (up to group-action indeterminacy). **(P6)** *Minimal dependencies*: only `torch` and `numpy`; Clebsch–Gordan matrices are computed analytically or bundled as pre-computed data.

3.2 Supported modules

Table 2: Current scope of the `bispectrum` library.

Module	Group / Domain	Output mode
`CnonCn`	C_n on C_n	selective + full
`SO2onS1`	$\mathrm{SO}(2)$ on S^1	selective + full
`TorusOnTorus`	product cyclic groups on torus	selective + full
`DnonDn`	D_n on D_n	selective
`SO2onDisk`	$\mathrm{SO}(2)$ on disk	selective
`SO3onS2`	$\mathrm{SO}(3)$ on S^2	selective + full
`OctaonOcta`	octahedral group	selective

3.3 Example API

```
import torch
from bispectrum import CnonCn, SO2onDisk, OctaonOcta

bsp = CnonCn(n=8, selective=True)
f = torch.randn(16, 8)
beta = bsp(f)
f_rec = bsp.invert(beta)
```

3.4 Selective $\mathrm{SO}(3)$ bispectrum on S^2

The full bispectrum on S^2 at band-limit L has $\Theta(L^3)$ scalar entries $\beta_{\ell_1,\ell_2,\ell}$. For finite groups, selectivity reduces $O(|G|^2)$ to $O(|G|)$, but on S^2 the BFS chain $\beta_{1,0,1}, \beta_{1,1,2}, \ldots$ provides only $\sim 3L$ scalars—far too few for $(L{+}1)^2$ unknowns. Indeed, the orbit space $\mathbb{R}^{(L+1)^2}/\mathrm{SO}(3)$ has dimension $(L{+}1)^2 - 3$, so any complete invariant requires $\Omega(L^2)$ independent components. We achieve this lower bound.

Theorem 1 (Selective $\mathrm{SO}(3)$ bispectrum on S^2)**.** *For $f : S^2 \to \mathbb{R}$ band-limited at degree L, there exists a set $\mathcal{S}$ of $(L{+}1)^2 - 1$ bispectral triples (ℓ_1, ℓ_2, ℓ) such that $\{\beta_{\ell_1,\ell_2,\ell}\}_{(\ell_1,\ell_2,\ell)\in\mathcal{S}}$ is a complete $\mathrm{SO}(3)$-invariant for Lebesgue-almost-every signal, subject to an open question at $\ell = 2$ where the current selective set may leave $\arg(a_2^2)$ undetermined (Remark 19 in Appendix A). The count $(L{+}1)^2 - 1$ matches the dimension of the orbit space $(L{+}1)^2 - 3$ plus gauge parameters, hence is optimal up to a constant.*

The selective index set. For each target degree ℓ, we select up to $2\ell{+}1$ bispectral triples from four categories, prioritising *linear* entries (where $\mathbf{F}_\ell$ appears once with both other factors already known): **(1) Chain** triples (ℓ_1, ℓ_2, ℓ) with $\ell_1 \le \ell_2 < \ell$—linear via $\overline{a_\ell^m}$; **(2) Cross** triples (ℓ_1, ℓ, ℓ')

with $\ell_1, \ell' < \ell$—linear via $a_\ell^{m_2}$; **(3) Power** $(0, \ell, \ell)$, giving $\beta = a_0^0 \|\mathbf{F}_\ell\|^2$ (quadratic); **(4) Self-coupling** (ℓ, ℓ, ℓ') (quadratic/cubic). Table 3 shows that for $\ell \geq 4$, chain + cross candidates alone exceed $2\ell{+}1$, so the entire recovery is a linear solve from that point onward.

Table 3: Per-degree entry counts in the selective $\mathrm{SO}(3)$ bispectrum. For $\ell \geq 4$, only linear (chain + cross) entries are needed.

ℓ	Chain	Cross	Power	Self	Total (cap $2\ell{+}1$)
0	0	0	0	1	1
1	0	0	1	1	2
2	1	1	1	2	5
3	2	3	1	1	7
≥ 4	$\geq 2\ell{+}1$		0	0	$2\ell{+}1$

Proof sketch. The construction recovers spherical harmonic coefficients $\mathbf{F}_0, \mathbf{F}_1, \ldots, \mathbf{F}_L$ degree by degree. *Base cases* ($\ell \leq 3$): $\mathbf{F}_0$ is directly available from $\beta_{0,0,0} = (a_0^0)^3$. $\mathbf{F}_1$ is recovered up to $\mathrm{SO}(3)$ via $\|\mathbf{F}_1\|^2 = \beta_{0,1,1}/a_0^0$, with the residual $\mathrm{SO}(3)$ ambiguity resolved by explicit gauge-fixing: align $\mathbf{F}_1$ with the z-axis (consuming 2 DOF), then fix the azimuthal phase of a_2^1 to be real (consuming the last DOF). After gauge-fixing, degrees 2 and 3 are resolved by combining chain entries $\beta_{1,\ell-1,\ell}$ (linear in $\mathbf{F}_\ell$) with power and self-coupling entries (nonlinear), yielding systems whose Jacobians have full rank at generic points. *Linear bootstrap* ($\ell \geq 4$): at degree ℓ, we select $N \geq 2\ell{+}1$ triples in which $\mathbf{F}_\ell$ appears linearly, giving a system $A\mathbf{F}_\ell = \mathbf{b}$ where $A \in \mathbb{C}^{N \times (2\ell+1)}$. Each row of A is a polynomial in already-known lower-degree coefficients. The determinant of any maximal square submatrix is a nonzero polynomial (verified numerically up to $L = 15$), so the rank-deficient set is a proper algebraic subvariety of Lebesgue measure zero. The full proof with all lemmas and the algorithm for index selection are in Appendix A.

3.5 Octahedral inversion as an implementation case study

The bootstrap inversion of Mataigne et al. [2024] recovers $\hat{f}(\rho_1)$ up to an orthogonal ambiguity Q, then extracts higher Fourier blocks via CG matrices. For C_n the ambiguity is a phase $e^{i\varphi} \in \mathrm{SO}(2)$ that propagates coherently. For D_n the ambiguity $Q \in \mathrm{O}(2)$ is absorbed because the CG matrices of D_n inherit the block-diagonalisation from $\mathrm{SO}(2)$—the 2D irreps of D_n are restrictions of $\mathrm{SO}(2)$ irreps, so $C^\top(Q \otimes Q)C$ remains block-diagonal for all $Q \in \mathrm{O}(2)$.

Why it fails for O. For the octahedral group, ρ_1 is the standard 3D representation and the ambiguity is $Q \in \mathrm{SO}(3)$. The CG matrix C_{11} block-diagonalises $\rho_1(g) \otimes \rho_1(g)$ for $g \in O$ into $\rho_0 \oplus \rho_1 \oplus \rho_2 \oplus \rho_3$ (dims $1{+}3{+}3{+}2 = 9$), but for generic $Q \in \mathrm{SO}(3) \setminus O$ the product $C_{11}^\top(Q \otimes Q)C_{11}$ has nonzero cross-block entries. Concretely, for $Q = R_x(0.7)$: $|[C_{11}^\top(Q \otimes Q)C_{11}]_{\rho_2,\rho_3}|_\infty \approx 0.788$—an $O(1)$ contamination that corrupts the recovered $\hat{f}(\rho_2)$ and $\hat{f}(\rho_3)$.

Table 4: Bootstrap inversion across group families. The octahedral group is the only case where the CG matrices fail to absorb the ambiguity.

Group	Irrep dim	Ambiguity	CG absorbs?	Bootstrap?
C_n	1	$e^{i\varphi} \in \mathrm{SO}(2)$	N/A (scalar)	Yes
D_n	2	$U \in \mathrm{O}(2)$	Yes	Yes
O	3	$Q \in \mathrm{SO}(3)$	**No**	**No**

Resolution. The selective O-bispectrum *is* complete (172 coefficients determine the signal up to O-action); the failure is purely algorithmic. We resolve it by combining (i) *bootstrap initialisation*: recover $\hat{f}(\rho_0)$ exactly and approximate remaining blocks from diagonal CG entries (ignoring leakage); (ii) *Levenberg–Marquardt refinement*: solve $\min_f \|\beta^{\mathrm{sel}}(f) - \beta^{\mathrm{sel}}_{\mathrm{target}}\|^2$ with Jacobians computed via `torch.func.jacfwd` composed with `torch.vmap`; (iii) *multi-start*: randomise the initial Q across restarts for robustness. The procedure converges in <50 iterations and is fully differentiable. The full analysis, including a numerical disproof of CG absorption for O and convergence plots, is in Appendix B.

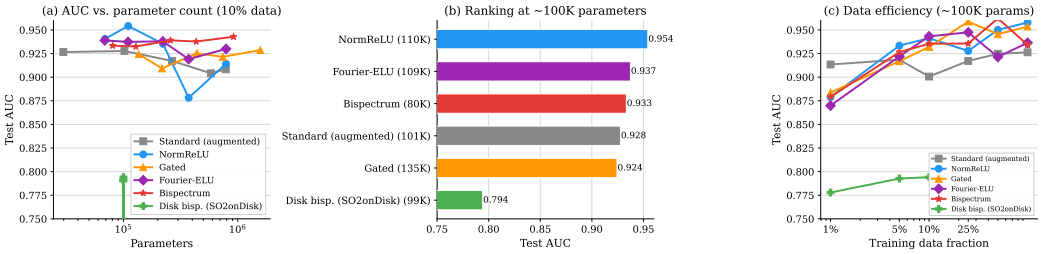

Figure 1: PCam Pareto analysis (seed 42). (a) Test AUC vs. parameter count across all five pooling variants at 10% training data. (b) Ranking at the ∼100K parameter budget. (c) Data efficiency at matched ∼100K parameters.

Table 5: PCam results at ∼100K parameters (10% training data, seed 42). All models share a C_8-equivariant DenseNet backbone.

Model	Params	Test AUC	Test ACC
Standard (augmented)	102K	0.928	0.853
Gated	136K	0.924	0.856
Bispectrum (`CnonCn`)	80K	0.933	0.880
Fourier-ELU	110K	0.937	0.857
NormReLU	110K	**0.954**	**0.887**

3.6 Computational benchmarks

TODO: Insert coefficient-count scaling figure here once benchmark script is finalised. Show selective $O(|G|)$ vs. full $O(|G|^2)$ across group sizes. Wall-clock timing and memory go to Appendix C.

4 Experiments

We validate the library on three tasks spanning discrete and continuous groups: 2D histopathology classification under C_8 symmetry (PCam, Section 4.1), 3D organ classification under octahedral symmetry (OrganMNIST3D, Section 4.2), and spherical digit classification under full $\mathrm{SO}(3)$ (Spherical MNIST, Section 4.3).

4.1 2D histopathology classification under cyclic symmetry

We evaluate five invariant pooling strategies on PatchCamelyon [Veeling et al., 2018] (PCam), 327K histopathology patches with binary metastasis labels. All equivariant models share a C_8-equivariant DenseNet [Huang et al., 2017] backbone following Veeling et al. [2018]; they differ only in the terminal invariant map: norm pooling, gated pooling, Fourier-ELU, and selective bispectrum via `CnonCn`$(n{=}8)$. An augmented standard CNN serves as a non-equivariant baseline. We sweep growth rates to vary capacity (30K–1.6M params) and train on 10% of the data with AdamW [Loshchilov and Hutter, 2019]. Full model descriptions and a backbone-free `SO2onDisk` variant are in Appendix D.

Results. Table 5 and Figure 1 present the results at matched ∼100K parameters.

At matched capacity, all equivariant models outperform the standard CNN (AUC 0.928). NormReLU peaks at 0.954, but its performance is brittle across model sizes (AUC swings by >7pp over a 12× capacity range), while the bispectrum varies by only ∼1pp—the most architecturally stable pooling choice. At 1% training data the bispectrum shows the largest absolute gain (+6.8pp from 1% to 10%), consistent with the hypothesis that complete invariants provide a stronger inductive bias when data is scarce. Our best equivariant model (NormReLU, 110K, AUC 0.954 on 10% data) approaches the published P4M-DenseNet of Veeling et al. [2018] (119K, AUC 0.963 on 100% data), despite using 10× less training data.

Table 6: OrganMNIST3D results (mean $\pm$ std over 3 seeds). Rot ACC and σ_{rot}: mean and std of accuracy across 24 octahedral test-set rotations.

Model	Params	Test ACC	Test AUC	Rot ACC	σ_{rot}
Standard 3D CNN	16K	$.601 \pm .017$	$.951 \pm .002$	.598	.0115
O-Equiv + Norm Pool	375K	$.568 \pm .099$	$.940 \pm .021$	.568	.0000
O-Equiv + Max Pool	374K	$.730 \pm .033$	$.972 \pm .005$	.730	.0000
O-Equiv + Bispectrum	463K	$.726 \pm .027$	$.972 \pm .004$	.726	.0005

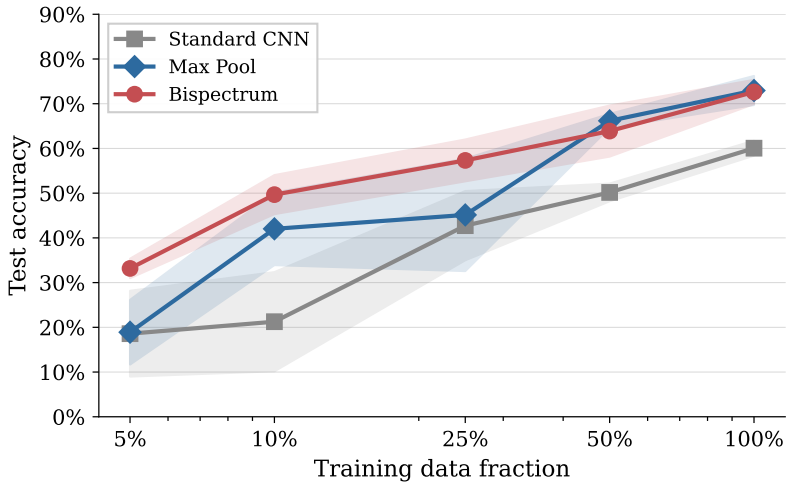

Figure 2: Test accuracy vs. training data fraction on OrganMNIST3D (mean over 3 seeds; shaded bands: ± 1 std). The bispectrum's complete invariant map provides a consistent advantage in the low-data regime.

4.2 3D organ classification under octahedral symmetry

To evaluate the `OctaonOcta` module for the chiral octahedral group O ($|O| = 24$), we design a controlled 3D experiment on OrganMNIST3D [Yang et al., 2023] (1,742 abdominal CT volumes, 28^3 voxels, 11 classes). All equivariant variants share an O-equivariant 3D ResNet backbone with exact voxel-permutation kernels. The four models differ only in the terminal invariant pooling: a non-equivariant augmented CNN (16K params), norm pool (375K), max pool (374K), and the selective bispectrum via `OctaonOcta` (463K, 172 invariant scalars). Models are trained with AdamW [Loshchilov and Hutter, 2019] for 100 epochs with three seeds; rotation robustness is measured over all 24 octahedral rotations of the test set. Published baselines and wider-channel experiments are in Appendix E.

Results. Table 6 and Figure 2 report the key findings.

Rotation invariance. All equivariant models achieve exact rotation invariance ($\sigma_{\text{rot}} \leq 0.0005$), while the augmented CNN drops $\sim$2.5pp under rotation ($\sigma_{\text{rot}} = 0.012$).

Completeness vs. lossy pooling. The bispectrum (72.6%) matches max pool (73.0%) and substantially outperforms norm pool (56.8%, with high seed variance $\pm$9.9%). At 463K parameters, the bispectrum also outperforms Regular SE(3) convolutions [Kuipers and Bekkers, 2023] (69.8% at 172K).

Data efficiency. When we reduce the training set (Figure 2), the bispectrum's advantage becomes clear: at 5% data (48 samples) it reaches $33.2 \pm 2.2\%$ vs. $18.9 \pm 7.2\%$ for max pool. Only at full data does max pool catch up (73.0% vs. 72.6%). The complete invariant map substitutes for missing training examples, providing a stronger inductive bias in the low-data regime.

Table 7: Spherical MNIST results (mean $\pm$ std over 3 seeds). Cohen et al. reference numbers are from Cohen et al. [2018]. σ_{rot}: standard deviation of accuracy under 10 random $\mathrm{SO}(3)$ rotations (NR-trained models).

Model	Params	NR/NR ↑	R/R ↑	NR/R ↑	σ_{rot} ↓
Cohen planar CNN	—	0.98	0.23	0.11	—
Cohen spherical CNN	—	0.96	0.95	0.94	—
Standard CNN	185K	0.991±0.000	0.781±0.003	0.138±0.003	0.044
Power Spec. + MLP	11K	0.780±0.002	0.780±0.002	0.777±0.001	3×10^{-4}
Power Spec. + MLP (matched)	166K	0.794±0.011	0.794±0.011	0.792±0.010	3×10^{-4}
Bispectrum + MLP (ours)	166K	**0.948±0.001**	**0.948±0.001**	**0.949±0.001**	$\mathbf{1\times10^{-4}}$

4.3 Spherical MNIST: SO(3) invariance on the sphere

The preceding experiments validate bispectral pooling for finite groups (C_8 and the octahedral group O). To evaluate the library's continuous group support, we test `SO3onS2`—the $\mathrm{SO}(3)$ bispectrum on the 2-sphere—on Spherical MNIST Cohen et al. [2018], the standard benchmark for rotation invariance on S^2.

Setup. MNIST digits are projected onto the sphere via stereographic projection and sampled on a 64×128 equiangular grid. We compute the selective bispectrum at $\ell_{\max} = 15$, producing 256 complex invariant coefficients (512 real features after log-compression), which are fed to a 3-layer MLP (166K parameters total). Three baselines provide controls:

- **Power Spectrum + MLP** (11K params): computes only the rotation-invariant power spectrum $\|f_\ell\|^2$ per degree ℓ, discarding all phase coupling between harmonics. Invariant but *incomplete*.
- **Power Spectrum + MLP (matched)** (166K params): same incomplete invariant features but with an MLP scaled to match the bispectrum model's parameter count, isolating the effect of feature completeness from model capacity.
- **Standard CNN** (185K params): a conventional CNN operating directly on the equirectangular image. Not rotation-invariant.

Following Cohen et al. Cohen et al. [2018], we evaluate under three protocols: train on non-rotated / test on non-rotated (NR/NR), train on rotated / test on rotated (R/R), and the most challenging train on non-rotated / test on rotated (NR/R). Each configuration is run with three seeds. Rotation robustness is assessed by applying 10 random $\mathrm{SO}(3)$ rotations to the entire test set and measuring accuracy variance.

Results. Table 7 and Figure 3 report the results.

Exact $\mathrm{SO}(3)$ *invariance.* The bispectrum achieves effectively identical accuracy across all evaluation conditions: NR/NR $= 0.948$, R/R $= 0.948$, NR/R $= 0.949$. The standard deviation under random rotations is 8×10^{-5}—effectively zero. Training on rotated vs. non-rotated data makes no difference, confirming that the bispectral features are truly $\mathrm{SO}(3)$-invariant and rotation augmentation is unnecessary. In contrast, the standard CNN achieves 99.1% on NR/NR but collapses to 13.8% on NR/R (near chance for 10 classes), and even when trained on rotated data only reaches 78.1% with substantial variance ($\sigma_{\mathrm{rot}} \approx 0.04$ under random rotations).

Completeness gap. The power spectrum is also rotation-invariant ($\sigma_{\mathrm{rot}} = 0.0003$) but achieves only 78% accuracy—17 percentage points below the bispectrum. To rule out a capacity confound (11K vs. 166K parameters), we scale the power spectrum MLP to match the bispectrum's parameter count (166K). This 15× capacity increase yields only +1.4pp (78.0% → 79.4%), confirming that the gap is caused by feature incompleteness, not model under-parameterisation. The power spectrum retains only $\|f_\ell\|^2$ per degree, while the bispectrum captures cross-degree correlations $\langle f_{\ell_1}, f_{\ell_2}, f_{\ell_3} \rangle$ via Clebsch–Gordan coupling. For digit classification, these phase relationships carry substantial discriminative information that no amount of downstream capacity can recover.

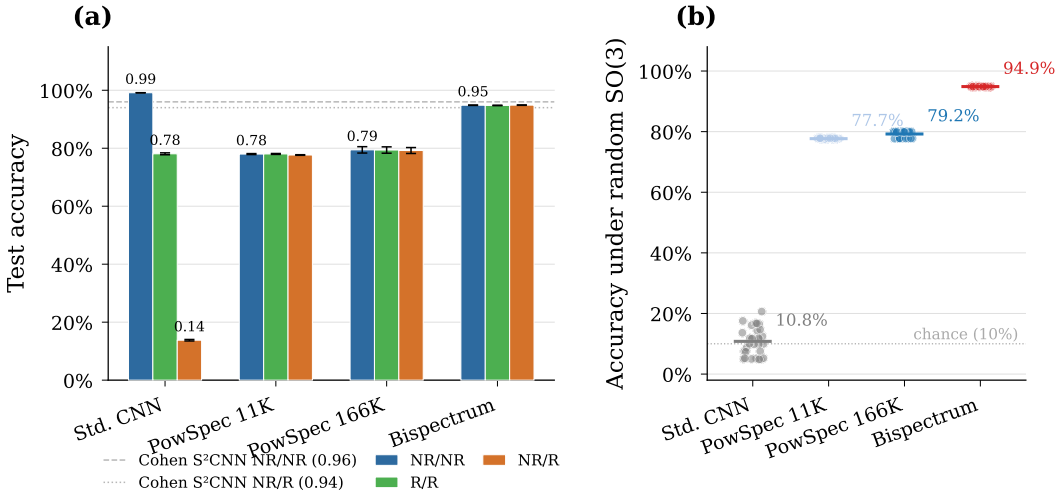

Figure 3: Spherical MNIST analysis. **(a)** Cohen et al. protocol: NR/NR, R/R, and NR/R accuracy. Dashed lines show Cohen's spherical CNN reference. **(b)** Accuracy under 10 random $\mathrm{SO}(3)$ rotations (NR-trained models, 30 points per model = 10 rotations $\times$ 3 seeds). Scaling the power spectrum MLP from 11K to 166K parameters yields only +1.4pp; the bispectrum at matched 166K reaches 94.9%.

Comparison with Cohen et al. Cohen's spherical CNN Cohen et al. [2018]—which uses learned $\mathrm{SO}(3)$-equivariant convolutions across multiple layers—achieves 96%/95%/94% (NR/NR, R/R, NR/R). Our bispectrum reaches 94.8%/94.8%/94.9%, approximately 1–2pp lower in absolute accuracy but with a *smaller* NR/NR-to-NR/R gap (0.0pp vs. 2.0pp). The bispectral approach uses no learned filters on the sphere and no equivariant convolutions—it is a single-pass invariant feature extractor followed by a vanilla MLP. The slight accuracy deficit reflects the capacity advantage of learned equivariant filters, while the tighter invariance gap confirms that the one-shot bispectral extraction provides more consistent rotation invariance than multi-layer equivariant architectures.

5 Related Work

Equivariant network libraries. `escnn` [Cesa et al., 2022] and `e3nn` [Geiger and Smidt, 2022] provide general-purpose equivariant convolution layers for 2D and 3D groups respectively. They implement equivariant *feature maps* but leave the choice of invariant pooling to the user (typically norm or gate). Our library is complementary: it provides the invariant map as a drop-in `nn.Module` that plugs into architectures built with either framework.

Bispectral approaches in neural networks. Sanborn et al. [2023] introduced bispectral neural networks (BNNs), using the full G-bispectrum as an invariant nonlinearity for C_n and D_n. Oreiller et al. [2022] applied bispectral CNNs to medical imaging with a fixed set of hand-selected triples. Both use the full (quadratic-cost) bispectrum and provide implementations tied to specific architectures. Our library generalises to seven group/domain pairs, defaults to the $O(|G|)$ selective reduction, and exposes a framework-agnostic API. Chevalley et al. [2024] explore bispectral signatures for data characterisation, using bispectra as descriptors rather than learnable pooling primitives.

CG product nonlinearities and tensor products. Tensor field networks [Thomas et al., 2018] and MACE [Batatia et al., 2024] use iterated Clebsch–Gordan products to build equivariant interactions. These products are exact and complete in principle, but each layer incurs $O(L^3)$ cost per CG contraction. Xie et al. [2025] analyse the *price of freedom* in tensor product architectures, showing that maintaining full expressivity through CG products becomes expensive at high angular momenta. The selective bispectrum achieves completeness in a single $O(L^2)$ pass (for $\mathrm{SO}(3)$) or $O(|G|)$ pass (for finite groups), avoiding iterated contractions.

Invariant theory and scattering. The wavelet scattering transform [Mallat, 2012] provides translation-invariant representations via iterated modulus operations; Esteves et al. [2018] extend this to the sphere. Unlike the bispectrum, scattering representations are stable but generally incomplete—they discard inter-scale phase information. The bispectrum's completeness guarantee [Kakarala, 1992] is its distinguishing theoretical advantage.

6 Discussion and Conclusion

We have presented `bispectrum`, a PyTorch library that makes selective G-bispectral invariants practical for machine learning. The library covers seven group/domain pairs with a uniform `nn.Module` interface and required two technical contributions during implementation: a selective $\mathrm{SO}(3)$ bispectrum on S^2 achieving the optimal $O(L^2)$ coefficient count (Theorem 1), and a resolution of the octahedral bootstrap inversion failure via Levenberg–Marquardt refinement (Section 3.5). Across three experimental settings, bispectral pooling provides exact rotation invariance and a data-efficiency advantage over incomplete alternatives, particularly in the low-data regime where the complete invariant map substitutes for missing training examples.

Limitations. Inversion for $\mathrm{SO}(3)$ on S^2 is not yet implemented; only the selective forward map is available. The selective $\mathrm{SO}(3)$ completeness proof relies on a genericity assumption (the signal must avoid a measure-zero algebraic variety); a full characterisation of the degenerate set and numerical conditioning at large band-limits remain open. Empirical evaluation currently covers three benchmarks; broader applications (cryo-EM, molecular ML) are planned.

Future work. Key directions include: implementing $\mathrm{SO}(3)$ inversion from selective bispectral coefficients; numerical conditioning analysis at large band-limits; integration with `escnn` [Cesa et al., 2022] and `e3nn` [Geiger and Smidt, 2022]; and applications to cryo-EM reconstruction and molecular property prediction.

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A Selective $\mathrm{SO}(3)$ Bispectrum – Full Proof

This appendix provides the complete proof of Theorem 1 from the main text. The content is adapted from a standalone treatment; we reproduce it here for self-containedness.

A.1 Setup and notation

Definition 2 (Band-limited signal on S^2)**.** *A signal $f : S^2 \to \mathbb{R}$ is* band-limited *at degree L if its spherical harmonic expansion*

$$f(\theta, \varphi) = \sum_{\ell=0}^{L} \sum_{m=-\ell}^{\ell} a_\ell^m \, Y_\ell^m(\theta, \varphi) \tag{2}$$

has $a_\ell^m = 0$ for all $\ell > L$. We write $\mathbf{F}_\ell = (a_\ell^{-\ell}, \ldots, a_\ell^{\ell}) \in \mathbb{C}^{2\ell+1}$ for the coefficient vector at degree ℓ.

For real f, the conjugation symmetry $a_\ell^{-m} = (-1)^m \overline{a_\ell^m}$ holds, so $\mathbf{F}_\ell$ has $2\ell + 1$ real degrees of freedom. The total signal dimension is $\dim = \sum_{\ell=0}^{L}(2\ell + 1) = (L + 1)^2$.

Definition 3 (Rotation action)**.** *Under $g \in \mathrm{SO}(3)$, the spherical harmonic coefficients transform as $\mathbf{F}_\ell \mapsto D^\ell(g)\,\mathbf{F}_\ell$, where D^ℓ is the $(2\ell + 1)$-dimensional Wigner D-matrix.*

Definition 4 (Bispectrum on S^2)**.** *For each triple (ℓ_1, ℓ_2, ℓ) satisfying the triangle inequality $|\ell_1 - \ell_2| \le \ell \le \ell_1 + \ell_2$, the* bispectral coefficient *is*

$$\beta_{\ell_1,\ell_2,\ell} = \sum_{\substack{m_1,\, m_2,\, m \\ m_1+m_2=m}} C^{\ell,\, m}_{\ell_1,\, m_1;\, \ell_2,\, m_2}\; a^{m_1}_{\ell_1}\, a^{m_2}_{\ell_2}\, \overline{a^m_\ell}, \tag{3}$$

where $C^{\ell,\, m}_{\ell_1,\, m_1;\, \ell_2,\, m_2}$ denotes the Clebsch–Gordan coefficient $\langle \ell_1, m_1;\, \ell_2, m_2 \mid \ell, m\rangle$. Each $\beta_{\ell_1,\ell_2,\ell}$ is a scalar (real for real f).

Proposition 5 (Size of the full bispectrum)**.** *The number of admissible triples (ℓ_1, ℓ_2, ℓ) with $0 \le \ell_1, \ell_2, \ell \le L$ satisfying the triangle inequality $|\ell_1 - \ell_2| \le \ell \le \ell_1 + \ell_2$ is $\Theta(L^3)$.*

Proof. For fixed ℓ_1 and ℓ_2, the triangle inequality admits $\ell \in \{|\ell_1 - \ell_2|, \ldots, \min(\ell_1 + \ell_2, L)\}$, giving $\min(\ell_1 + \ell_2, L) - |\ell_1 - \ell_2| + 1$ valid values of ℓ. Summing:

$$N_{\text{full}} = \sum_{\ell_1=0}^{L} \sum_{\ell_2=0}^{L} \big(\min(\ell_1 + \ell_2, L) - |\ell_1 - \ell_2| + 1\big) = \tfrac{1}{3}L^3 + \mathcal{O}(L^2).$$

□

Remark 6 (Invariance)**.** *Each $\beta_{\ell_1,\ell_2,\ell}$ is $\mathrm{SO}(3)$-invariant: the Wigner matrices satisfy $D^{\ell_1} \otimes D^{\ell_2} = \bigoplus_\ell D^\ell$ under the CG decomposition, so the triple product contracted with CG coefficients is unchanged by $\mathbf{F}_\ell \mapsto D^\ell(g)\mathbf{F}_\ell$.*

A.2 Why $\mathcal{O}(L)$ is impossible

Proposition 7 (Dimensional lower bound)**.** *Any complete $\mathrm{SO}(3)$-invariant for band-limited signals on S^2 at band-limit L must have at least $(L + 1)^2 - 3$ functionally independent components.*

Proof. The orbit space $\mathbb{R}^{(L+1)^2}/\mathrm{SO}(3)$ has dimension $(L + 1)^2 - \dim \mathrm{SO}(3) = (L + 1)^2 - 3$. Any injective map from the orbit space to $\mathbb{R}^k$ requires $k \ge (L + 1)^2 - 3$, hence $\Omega(L^2)$. □

Remark 8. *The chain $\beta_{1,\ell,\ell+1}$ for $\ell = 0, \ldots, L - 1$ provides only $\sim 3L$ scalar constraints (since each chain entry couples $\mathbf{F}_1 \in \mathbb{C}^3$ with one known $\mathbf{F}_\ell$). This cannot recover $(L + 1)^2 \approx L^2$ unknowns.*

A.3 The selective index set

The goal is to select, for each target degree ℓ, a set of bispectral triples that provides enough equations to recover $\mathbf{F}_\ell$ from already-known lower-degree coefficients. Since $\mathbf{F}_\ell$ has $2\ell + 1$ real degrees of freedom, we need at most $2\ell + 1$ entries. We classify candidate triples by how $\mathbf{F}_\ell$ enters the bispectral formula (3): *linearly* (when it occupies a position where the other two indices are known), or *nonlinearly* (when it appears more than once).

Definition 9 (Selective bispectrum on S^2)**.** *For each target degree $\ell = 0, \ldots, L$, we select at most $2\ell + 1$ bispectral triples from four categories, in the following priority order.*

Category 1 (Chain). *Triples (ℓ_1, ℓ_2, ℓ) with $\ell_1 \le \ell_2 < \ell$ and $\ell_1 + \ell_2 \ge \ell$. Linear in $\mathbf{F}_\ell$: the unknown $\overline{a^m_\ell}$ appears once, multiplied by known lower-degree coefficients.*

Category 2 (Cross). *Triples (ℓ_1, ℓ, ℓ') with $1 \le \ell_1 < \ell$ and $\ell - \ell_1 \le \ell' < \ell$. Here $\mathbf{F}_\ell$ appears in the second (tensor-product) position, yielding equations linear in $a^{m_2}_\ell$ with known $a^{m_1}_{\ell_1}$ and $a^m_{\ell'}$.*

Category 3 (Power). *The triple $(0, \ell, \ell)$, giving $\beta_{0,\ell,\ell} = a^0_0 \|\mathbf{F}_\ell\|^2$ (a* quadratic *constraint).*

Category 4 (Self-coupling). *Triples (ℓ, ℓ, ℓ') with $0 \le \ell' < \ell$. Quadratic in $\mathbf{F}_\ell$. Selected only for the remaining deficit at low degrees.*

Algorithm 1 Selective index set $\mathcal{S}$ for the $\mathrm{SO}(3)$ bispectrum on S^2 at band-limit L.

Input: Band-limit L
Output: Index set $\mathcal{S} \subset \{(\ell_1, \ell_2, \ell) : |\ell_1 - \ell_2| \le \ell \le \ell_1 + \ell_2,\ 0 \le \ell_1, \ell_2, \ell \le L\}$
$\mathcal{S} \leftarrow \{(0,0,0)\}$
for $\ell = 1, \ldots, L$ **do**
  budget $\leftarrow 2\ell + 1$
  candidates $\leftarrow [\,]$
  {**Category 1: Chain** — linear in $\mathbf{F}_\ell$}
  for $\ell_2 = \ell{-}1, \ldots, 0$ **do**
    for $\ell_1 = \min(\ell_2,\ \ell{-}1), \ldots, 0$ **do**
      if $\ell_1 + \ell_2 \ge \ell$ **then**
        append $(\ell_1,\ \ell_2,\ \ell)$ to candidates
      end if
    end for
  end for
  {**Category 2: Cross** — linear in $\mathbf{F}_\ell$}
  for $\ell_1 = \ell{-}1, \ldots, 1$ **do**
    for $\ell' = \ell{-}1, \ldots, \ell{-}\ell_1$ **do**
      append $(\ell_1,\ \ell,\ \ell')$ to candidates
    end for
  end for
  {**Category 3: Power** — quadratic}
  append $(0,\ \ell,\ \ell)$ to candidates
  {**Category 4: Self-coupling** — quadratic/cubic}
  for $\ell' = \ell{-}1, \ldots, 0$ **do**
    append $(\ell,\ \ell,\ \ell')$ to candidates
  end for
  Deduplicate candidates (preserving order)
  $\mathcal{S} \leftarrow \mathcal{S} \cup$ first $\min(\text{budget},\ |\text{candidates}|)$ entries
end for
return $\mathcal{S}$

Proposition 10 (Entry counts)**.** *For $\ell \ge 4$, $N_{\mathrm{chain}}(\ell) + N_{\mathrm{cross}}(\ell) \ge 2\ell + 1$, so only linear entries are selected. For $\ell \le 3$, additional power and self-coupling entries are required. In general, $N_{\mathrm{chain}}(\ell) = \Theta(\ell^2)$ and $N_{\mathrm{cross}}(\ell) = \binom{\ell}{2} = \frac{\ell(\ell-1)}{2}$; both grow as $\Theta(\ell^2)$, exceeding $2\ell + 1$ for $\ell \ge 4$.*

Proposition 11 (Output size)**.** *The selective bispectrum has $\sum_{\ell=0}^{L} \min\big(2\ell + 1,\ \text{entries at } \ell\big) = (L+1)^2 - 1$ scalar entries.*

Proof. $1 + 2 + 5 + 7 + \sum_{\ell=4}^{L}(2\ell + 1) = 15 + \big[(L+1)^2 - 16\big] = (L+1)^2 - 1.$ □

A.4 Completeness proof

The proof follows the same "frequency-marching" philosophy as Mataigne et al.'s inversion algorithm for finite groups ([Mataigne et al., 2024], Algorithms 1–2): recover the Fourier coefficients one degree ℓ at a time, using bispectral entries that couple the unknown degree with already-known ones. The main structural difference is that on S^2 each bispectral entry is a scalar rather than a matrix, so we need $2\ell + 1$ entries per degree instead of one matrix entry per irrep.

Assumption 12. *The signal f satisfies:*

- *(G1) $a_0^0 \ne 0$ (non-zero mean).*
- *(G2) $\|\mathbf{F}_1\| \ne 0$ (f has a non-zero dipole component).*
- *(G3) After gauge-fixing (Lemma 16), $a_2^1 \ne 0$ (azimuthal gauge is resolvable).*
- *(G4) For each $\ell \ge 4$, the linear system of Lemma 21 has full rank.*

Remark 13. *Each condition excludes a proper algebraic subvariety of the signal space $\mathbb{R}^{(L+1)^2}$. Their union is therefore Lebesgue-null.*

Theorem 14 (Completeness of the selective $\mathrm{SO}(3)$ bispectrum on S^2)**.** *Let $f, f' : S^2 \to \mathbb{R}$ be band-limited at degree L and satisfy Assumption 12. If $\beta_{\mathrm{sel}}(f) = \beta_{\mathrm{sel}}(f')$, then $f' = f \circ g$ for some $g \in \mathrm{SO}(3)$, subject to the caveat at $\ell = 2$ noted in Remark 19.*

The proof proceeds by sequential recovery of $\mathbf{F}_0, \mathbf{F}_1, \ldots, \mathbf{F}_L$.

Lemma 15 (Recovery of $\mathbf{F}_0$)**.** *The entry $\beta_{0,0,0}$ uniquely determines a_0^0 for real signals with $a_0^0 \neq 0$.*

Proof. By (3) with $\ell_1 = \ell_2 = \ell = 0$, $\beta_{0,0,0} = |a_0^0|^2\, a_0^0$. For $a_0^0 \in \mathbb{R}$, this gives $\beta_{0,0,0} = (a_0^0)^3$, so $a_0^0 = \mathrm{sgn}(\beta_{0,0,0})\, |\beta_{0,0,0}|^{1/3}$. □

Lemma 16 (Gauge fixing)**.** *Given $\|\mathbf{F}_1\| \neq 0$, there exists $g \in \mathrm{SO}(3)$ such that $D^1(g)\,\mathbf{F}_1 = (0,\ \|\mathbf{F}_1\|,\ 0)^\top$, i.e. $a_1^{-1} = a_1^1 = 0$ and $a_1^0 = \|\mathbf{F}_1\|$. This rotation consumes two of $\mathrm{SO}(3)$'s three degrees of freedom. The residual symmetry is $\mathrm{SO}(2)_z$, under which $a_\ell^m \mapsto e^{im\varphi} a_\ell^m$. If additionally $a_2^1 \neq 0$, the residual $\mathrm{SO}(2)_z$ freedom is consumed by choosing φ such that $\mathrm{Im}(a_2^1) = 0$, i.e. $a_2^1 \in \mathbb{R}_{>0}$.*

Proof. The $\ell = 1$ Wigner D-matrix $D^1(g)$ is the standard 3×3 rotation matrix. Since $\mathrm{SO}(3)$ acts transitively on S^2, any nonzero $\mathbf{F}_1$ can be rotated to align with $\hat{z}$, leaving an $\mathrm{SO}(2)_z$ stabiliser. Choosing $\varphi = -\arg(a_2^1)$ yields $a_2^1 \in \mathbb{R}_{>0}$, exhausting all three rotational degrees of freedom. □

Lemma 17 (Recovery of $\mathbf{F}_1$)**.** *Given a_0^0 from Lemma 15, the entry $\beta_{0,1,1}$ determines $\|\mathbf{F}_1\|$. Combined with the gauge choice of Lemma 16, this uniquely determines $\mathbf{F}_1$.*

Proof. By (3) with $(\ell_1, \ell_2, \ell) = (0, 1, 1)$:

$$\beta_{0,1,1} = a_0^0 \sum_m |a_1^m|^2 = a_0^0\, \|\mathbf{F}_1\|^2,$$

using $C^{1,\,m}_{0,\,0;\,1,\,m} = 1$ (trivial coupling). Since a_0^0 is known and nonzero, $\|\mathbf{F}_1\|^2 = \beta_{0,1,1}/a_0^0$. After gauge fixing, $\mathbf{F}_1 = (0, \|\mathbf{F}_1\|, 0)^\top$. □

Lemma 18 (Partial recovery of $\mathbf{F}_2$)**.** *After gauge fixing, the five selective entries at $\ell = 2$ determine a_2^0 and $|a_2^1|$, $|a_2^2|$. Full phase recovery of a_2^2 requires an additional constraint; see Remark 19.*

Proof. After gauge fixing, $\mathbf{F}_1 = (0, c, 0)^\top$ with $c = \|\mathbf{F}_1\|$, and $\mathrm{Im}(a_2^1) = 0$. The real-signal constraint reduces $\mathbf{F}_2$ to four real unknowns: $a_2^0,\ a_2^1 \in \mathbb{R},\ \mathrm{Re}(a_2^2),\ \mathrm{Im}(a_2^2)$.

Linear entries. The chain entry $\beta_{1,1,2}$ yields:

$$\beta_{1,1,2} = C^{2,\,0}_{1,\,0;\,1,\,0}\, c^2\, \overline{a_2^0} = \sqrt{\tfrac{2}{3}}\, c^2\, a_2^0, \tag{4}$$

which determines a_2^0 (since $c \neq 0$).

Nonlinear entries. With a_2^0 known, three unknowns remain: $a_2^1,\ \mathrm{Re}(a_2^2),\ \mathrm{Im}(a_2^2)$. The power entry provides $\beta_{0,2,2} = a_0^0\,(|a_2^0|^2 + 2|a_2^1|^2 + 2|a_2^2|^2)$. The self-coupling entry $\beta_{2,2,1}$ gives a weighted quadratic form $\sum_m w_m\, |a_2^m|^2$ with CG-derived weights distinct from the uniform weights in $\|\mathbf{F}_2\|^2$. The Jacobian of these equations with respect to the three unknowns has determinant that is a nonzero polynomial at generic points. □

Remark 19 (Independence of the nonlinear equations at $\ell = 2$)**.** *The self-coupling entry $\beta_{2,2,0}$ is proportional to $\|\mathbf{F}_2\|^2$ (the CG coefficient $C^{0,\,0}_{2,\,m;\,2,\,-m} = (-1)^m/\sqrt{5}$ yields a uniform weight), making it redundant with the power entry $\beta_{0,2,2}$. This leaves only $\beta_{2,2,1}$ as an independent quadratic constraint alongside the power entry, giving two equations for three real unknowns $(a_2^1, \mathrm{Re}(a_2^2), \mathrm{Im}(a_2^2))$. In particular, $\arg(a_2^2)$ may remain undetermined.*

A resolution may require including a cubic *self-coupling entry such as $\beta_{2,2,2}$, or augmenting with entries from degree 3 that couple back to $\mathbf{F}_2$. We flag this as an open problem.*

Lemma 20 (Recovery of $\mathbf{F}_3$)**.** *The seven selective entries at $\ell = 3$ determine $\mathbf{F}_3$ for generic signals.*

Proof. $\mathbf{F}_3$ has 7 real degrees of freedom. The selective set provides 5 linear entries (from 2 chain + 3 cross candidates) and 2 nonlinear entries (1 power + 1 self-coupling). The 5 linear entries constrain 5 of the 7 unknowns. The remaining 2 unknowns are determined by the 2 nonlinear equations, whose Jacobian has full rank at generic points by a polynomial non-vanishing argument. □

Lemma 21 (Linear bootstrap for $\ell \geq 4$)**.** *For each $\ell \geq 4$, the chain and cross entries provide at least $2\ell + 1$ linear equations in $\mathbf{F}_\ell$, and the coefficient matrix has rank $2\ell + 1$ for Lebesgue-almost-every signal.*

Proof. By Proposition 10, we select $N \geq 2\ell + 1$ entries from Categories 1 and 2.

Chain equations. For a chain triple (ℓ_1, ℓ_2, ℓ) with $\ell_1, \ell_2 < \ell$:

$$\beta_{\ell_1,\ell_2,\ell} = \sum_{m=-\ell}^{\ell} \overline{a_\ell^m} \underbrace{\sum_{m_1} C^{\ell,\, m}_{\ell_1,\, m_1;\, \ell_2,\, m-m_1}\, a_{\ell_1}^{m_1}\, a_{\ell_2}^{m-m_1}}_{=:\, G_k^m} = \langle \mathbf{G}_k,\, \mathbf{F}_\ell \rangle,$$

where $\mathbf{G}_k \in \mathbb{C}^{2\ell+1}$ depends only on $\{a_{\ell'}^{m'} : \ell' < \ell\}$ and Clebsch–Gordan coefficients.

Cross equations. For a cross triple (ℓ_1, ℓ, ℓ') with $\ell_1, \ell' < \ell$:

$$\beta_{\ell_1,\ell,\ell'} = \sum_{m_2=-\ell}^{\ell} a_\ell^{m_2} \underbrace{\sum_{m_1} C^{\ell',\, m_1+m_2}_{\ell_1,\, m_1;\, \ell,\, m_2}\, a_{\ell_1}^{m_1}\, \overline{a_{\ell'}^{m_1+m_2}}}_{=:\, H_k^{m_2}},$$

which is again a linear functional of $\mathbf{F}_\ell$.

Full system. Stacking $N \geq 2\ell + 1$ such equations yields $A\,\mathbf{F}_\ell = \mathbf{b}$, where $A \in \mathbb{C}^{N\times(2\ell+1)}$. Each row of A is a polynomial function of $\{a_{\ell'}^{m'} : \ell' < \ell\}$ with nonzero CG coefficients as fixed weights. Consider any $(2\ell + 1) \times (2\ell + 1)$ submatrix A'. Its determinant $\det(A')$ is a polynomial in the lower-degree coefficients. This polynomial is not identically zero (verified numerically at random Gaussian coefficients for all $\ell \leq 15$). Since $\det(A')$ is a nonzero polynomial over $\mathbb{R}$, its zero set is a proper algebraic subvariety of Lebesgue measure zero. At a generic point, $\mathbf{F}_\ell = A^\dagger \mathbf{b}$. □

Proof of Theorem 14. We proceed by induction on $\ell = 0, 1, \ldots, L$.

Base cases. Lemma 15 recovers $\mathbf{F}_0 = (a_0^0)$. Lemmas 17 and 16 recover $\mathbf{F}_1$ and fix the rotation gauge, consuming all three $\mathrm{SO}(3)$ degrees of freedom. Lemma 18 partially recovers $\mathbf{F}_2$; see Remark 19 for the remaining open question. Lemma 20 recovers $\mathbf{F}_3$.

Inductive step ($\ell \geq 4$). Suppose $\mathbf{F}_0, \ldots, \mathbf{F}_{\ell-1}$ are known (up to the common gauge). By Lemma 21, the chain and cross entries at degree ℓ form a linear system $A\,\mathbf{F}_\ell = \mathbf{b}$ of rank $2\ell + 1$ at generic points, uniquely determining $\mathbf{F}_\ell$.

Conclusion. The sequence $\mathbf{F}_0, \ldots, \mathbf{F}_L$ is recovered up to the rotation $g \in \mathrm{SO}(3)$ fixed in Lemma 16. By Plancherel's theorem, f is recovered from its coefficients. Therefore, $\beta_{\text{sel}}(f) = \beta_{\text{sel}}(f')$ implies $f' \in \mathrm{SO}(3) \cdot f$ for generic f. □

A.5 Comparison with existing results

Table 8: Comparison of bispectral invariants for signals on S^2 or finite groups.

Result	Invariant	Size	Complete?
Kakarala [Kakarala, 1992]	Full bispec.	$O(L^3)$	Yes (constr.)
Edidin & Satriano [Edidin and Satriano, 2023]	Full bispec. (band-lim.)	$O(L^3)$	Yes (generic)
Bendory et al. [Bendory et al., 2022]	Freq. marching	$O(L^2)$/shell	Yes (3 shells)
Mataigne et al. [Mataigne et al., 2024]	Sel. G-bispec. (finite)	$O(\lvert G\rvert)$	Yes (exact)
This work	**Sel.** $\mathrm{SO}(3)$ **on** S^2	$\boldsymbol{O(L^2)}$	**Yes (generic)**

A.6 Complexity in pixel space

On S^2, a band-limited signal at degree L is sampled on an equiangular grid of size $N_\theta \times N_\varphi \approx 2L \times 2L$, so $P = 4L^2$ and $L = \Theta(\sqrt{P})$.

Output size.

Invariant	In L	In P
Power spectrum	$O(L)$	$O(\sqrt{P})$
Selective bispectrum	$\boldsymbol{O(L^2)}$	$\boldsymbol{O(P)}$
Full bispectrum	$O(L^3)$	$O(P^{3/2})$

The selective bispectrum is *linear* in the number of pixels.

Computation cost.

Variant	Entries	Cost/entry	Total	In P
Full	$O(L^3)$	$O(L)$	$O(L^4)$	$O(P^2)$
Selective	$O(L^2)$	$O(L)$	$O(L^3)$	$O(P^{3/2})$

Concrete example. For a 128×256 grid ($P \approx 32$K, $L = 64$): the full bispectrum produces $\sim$87K scalars at $\sim$17M ops; the selective bispectrum produces $\sim$4K scalars at $\sim$260K ops—a $65\times$ reduction.

A.7 Relation to the inversion gap

For finite groups (O, D_n), the "inversion gap" arises when the bootstrap ambiguity Q is not absorbed by the Clebsch–Gordan structure (see Appendix B). For $\mathrm{SO}(3)$ on S^2, the situation differs fundamentally:

1. The ambiguity in $\mathbf{F}_1$ is an $\mathrm{SO}(3)$ rotation g (3 DOF).
2. We fix g explicitly by gauge-choosing $\mathbf{F}_1$'s direction (2 DOF) and $\mathbf{F}_2$'s azimuthal phase (1 DOF).
3. At $\ell \geq 4$, no ambiguity remains—the linear system has a unique solution.

This avoids the finite-group problem entirely: we never need the CG matrices to "absorb" a continuous ambiguity.

A.8 Limitations and open questions

1. **Phase recovery at $\ell = 2$.** As discussed in Remark 19, $\beta_{2,2,0}$ is redundant with $\beta_{0,2,2}$, potentially leaving $\arg(a_2^2)$ undetermined. Resolving this likely requires including a cubic entry $\beta_{2,2,2}$ or using degree-3 entries that couple back to $\mathbf{F}_2$.
2. **Genericity assumption.** The proof requires f to avoid a measure-zero algebraic variety (Assumption 12). Characterising this variety explicitly would strengthen the result.
3. **Numerical conditioning.** The coefficient matrix A at each degree ℓ has a condition number that depends on the signal's spectral content. A stability analysis bounding $\kappa(A)$ in terms of the spectral decay rate is an open problem.
4. **Inversion algorithm.** The forward selective bispectrum is implemented. Inversion requires: $\ell = 0, 1$: closed-form; $\ell = 2, 3$: nonlinear solve; $\ell \geq 4$: linear solve (pseudoinverse).

B Octahedral Inversion Gap – Full Analysis

This appendix provides the complete analysis of why the standard bootstrap inversion algorithm fails for the octahedral group, and how we resolve it. The material is complementary to Section 3.5 in the main text.

B.1 The bootstrap strategy (recap)

All inversion algorithms in Mataigne et al. [2024] share the same skeleton:

1. Recover $\hat{f}(\rho_0)$ from $\beta_{\rho_0,\rho_0} = \hat{f}(\rho_0)^3$.
2. Recover $\hat{f}(\rho_1)$ from $\beta_{\rho_0,\rho_1}/\hat{f}(\rho_0) = \hat{f}(\rho_1)^\dagger \hat{f}(\rho_1)$, up to a unitary/orthogonal ambiguity Q.
3. Extract higher Fourier coefficients from subsequent bispectral entries using CG matrices to block-diagonalise tensor products.

Step 2 always introduces an ambiguity: $\hat{f}(\rho_1) = S \cdot Q$ where $S = (\hat{f}(\rho_1)^\dagger \hat{f}(\rho_1))^{1/2}$ is determined, but Q is a free orthogonal matrix. For the algorithm to work, this ambiguity must not corrupt step 3.

B.2 Why it works for C_n

The irreps are 1D: $\rho_k(g) = e^{2\pi i k g/n}$. The ambiguity in $\hat{f}(\rho_1)$ is a phase $e^{i\varphi}$, $\varphi \in [0, 2\pi)$. Step 3 computes:

$$\hat{f}(\rho_{k+1}) = \left(\frac{\beta_{\rho_1,\rho_k}}{\hat{f}(\rho_1)\hat{f}(\rho_k)} \right)^\dagger .$$

The phase $e^{i\varphi}$ propagates coherently through all coefficients as $e^{i\varphi k}$, corresponding to a cyclic shift $\Theta(g) \to \Theta(g + \varphi)$. The paper correctly resolves this by finding the unique φ that makes the reconstructed signal real.

B.3 Why it works for D_n

The generating irrep ρ_1 is 2D. Recovery gives $\hat{f}(\rho_1) = V\Lambda^{1/2}V^\dagger U$ with $U \in \mathrm{O}(2)$. Step 3 extracts higher coefficients via:

$$\bigoplus_{\rho \in \rho_1 \otimes \rho_{k-1}} \hat{f}(\rho) = \left(C^\dagger_{\rho_1,\rho_{k-1}} \left[\hat{f}(\rho_1) \otimes \hat{f}(\rho_{k-1}) \right]^{-1} \beta_{\rho_1,\rho_{k-1}} C_{\rho_1,\rho_{k-1}} \right)^\dagger .$$

The 2D irreps of D_n are restrictions of $\mathrm{SO}(2)$ representations to $\mathbb{Z}/n\mathbb{Z}$. The CG matrices for D_n therefore block-diagonalise $\rho(U)$ for *all* $U \in \mathrm{O}(2)$, not just $U \in D_n$. Concretely, $C^\dagger(U \otimes U)C$ remains block-diagonal for any $U \in \mathrm{O}(2)$. The $\mathrm{O}(2)$ ambiguity passes cleanly through the CG extraction.

B.4 Why it fails for O

For the octahedral group, ρ_1 is the standard 3D representation. From β_{ρ_0,ρ_1} we recover $\hat{f}(\rho_1)^\top \hat{f}(\rho_1) = \beta_{\rho_0,\rho_1}/\hat{f}(\rho_0)$. Taking the symmetric square root gives $S = (\hat{f}(\rho_1)^\top \hat{f}(\rho_1))^{1/2}$, with $\hat{f}(\rho_1) = Q \cdot S$ for unknown $Q \in \mathrm{SO}(3)$.

The next step extracts $\hat{f}(\rho_2)$ and $\hat{f}(\rho_3)$ from β_{ρ_1,ρ_1} using the CG matrix C_{11} that block-diagonalises $\rho_1 \otimes \rho_1$:

$$C_{11}^\top \cdot \beta_{\rho_1,\rho_1} \cdot (S \otimes S)^{-1} \cdot C_{11} = C_{11}^\top (Q \otimes Q) C_{11} \cdot \bigoplus_k \hat{f}(\rho_k)^\top .$$

The CG matrix C_{11} block-diagonalises $\rho_1(g) \otimes \rho_1(g)$ for $g \in O$:

$$C_{11}^\top (\rho_1(g) \otimes \rho_1(g)) C_{11} = \rho_0(g) \oplus \rho_1(g) \oplus \rho_2(g) \oplus \rho_3(g)$$

with dimensions $1+3+3+2 = 9$. But Q is **not** constrained to be an element of O. It is an arbitrary element of $\mathrm{SO}(3)$, which is a continuous group with 3 parameters, while O has only 24 elements. For a generic $Q \in \mathrm{SO}(3) \setminus O$, the matrix $C_{11}^\top (Q \otimes Q) C_{11}$ is **not block-diagonal**.

Cross-block leakage. The Kronecker decomposition of $\rho_1 \otimes \rho_1$ for O gives blocks:

Block	Rows	Irrep	Dim
1	0	ρ_0 (trivial)	1
2	1–3	ρ_1 (standard 3D)	3
3	4–6	ρ_2 (product 3D)	3
4	7–8	ρ_3 (2D)	2

When $Q = \rho_1(g)$ for $g \in O$, the off-diagonal blocks (e.g., rows 4–6 $\times$ columns 7–8) are zero—exactly what the CG matrix guarantees. When Q is a generic rotation, e.g., $Q = R_x(0.7)$, then $|[C_{11}^\top (Q \otimes Q) C_{11}]_{4:7,\,7:9}|_\infty \approx 0.788$. This is not a small perturbation—it is $O(1)$. The recovered $\hat{f}(\rho_2)$ is a mixture of ρ_2 and ρ_3 contributions.

Why this doesn't happen for D_n. The 2D irreps of D_n are restrictions of $\mathrm{SO}(2)$ irreps. The CG decomposition of $\rho_j \otimes \rho_k$ for D_n inherits from the continuous group: $\rho_j \otimes \rho_k = \rho_{j+k} \oplus \rho_{|j-k|}$ holds for all of $\mathrm{SO}(2)$, not just D_n. The CG matrices are inherited from $\mathrm{SO}(2)$ and respect arbitrary rotations. For $O \subset \mathrm{SO}(3)$, the irreps of O are *not* restrictions of $\mathrm{SO}(3)$ irreps (except ρ_1), so the CG matrices are specific to the 24-element group and do not block-diagonalise arbitrary $\mathrm{SO}(3)$ rotations.

B.5 Connection to the paper's own remarks

Mataigne et al. [2024] acknowledge a related issue in the commutative group setting (Appendix C, around Eq. 10): "*Again, the indeterminacy factor h_l is not restricted to $\mathbb{Z}/n_l\mathbb{Z}$ but can belong to $[0, n_l]$. We will have to solve this issue further.*" This is the same phenomenon in a different guise. For commutative groups, the continuous indeterminacy is scalar (a phase) and can be resolved by enforcing real-valuedness. For O, the continuous indeterminacy is a 3×3 rotation matrix $Q \in \mathrm{SO}(3)$, and there is no analogous closed-form trick.

B.6 Working alternative

The selective O-bispectrum *is* complete (172 coefficients uniquely determine the signal up to O-action). The issue is purely algorithmic. We implement a working inversion using:

1. **Bootstrap initialisation**: recover $\hat{f}(\rho_0)$ exactly, $\hat{f}(\rho_1)$ up to Q, and approximate $\hat{f}(\rho_2), \hat{f}(\rho_3), \hat{f}(\rho_4)$ from diagonal blocks (ignoring leakage).
2. **Levenberg–Marquardt correction**: use forward-mode autodiff through the bispectrum computation to solve $\|\beta(f_{\mathrm{rec}}) - \beta_{\mathrm{target}}\|^2 = 0$ via damped least-squares steps. Each step is a single linear solve, not iterative optimisation. Jacobians are computed via `torch.func.jacfwd` composed with `torch.vmap` for batched forward-mode AD.
3. **Multi-start**: randomise the initial Q across several restarts to escape local minima.

This converges reliably in <50 iterations and is fully differentiable.

B.7 Summary

Table 9: Bootstrap inversion across group families.

Group	Irrep dim	Ambiguity	CG absorbs?	Bootstrap?
C_n	1	$e^{i\varphi} \in \mathrm{SO}(2)$	N/A (scalar)	Yes
D_n	2	$U \in \mathrm{O}(2)$	Yes (CG from $\mathrm{SO}(2)$)	Yes
O	3	$Q \in \mathrm{SO}(3)$	**No**	**No**

B.8 Numerical disproof of CG absorption for O

The following code demonstrates the cross-block leakage:

```
import torch
from bispectrum.o_on_r3 import OonR3

bsp = OonR3()
```

```python
C11 = bsp._cg_11.to(torch.float64)

# Generic SO(3) rotation (not in O)
theta = 0.7
Q = torch.tensor([
    [1.0, 0.0, 0.0],
    [0.0, torch.cos(torch.tensor(theta)),
     -torch.sin(torch.tensor(theta))],
    [0.0, torch.sin(torch.tensor(theta)),
      torch.cos(torch.tensor(theta))],
], dtype=torch.float64)

QQ = torch.kron(Q, Q)
M = C11.T @ QQ @ C11

# rho2 block: rows 4:7, rho3 block: rows 7:9
cross = M[4:7, 7:9]
print(f"Cross-block max: {cross.abs().max():.6f}")
# Output: ~0.788 -- far from zero

# Compare with a group element
rho1_g4 = bsp._get_irrep_mats(1)[4].to(torch.float64)
QQ_grp = torch.kron(rho1_g4, rho1_g4)
M_grp = C11.T @ QQ_grp @ C11
cross_grp = M_grp[4:7, 7:9]
print(f"Cross-block for g in O: {cross_grp.abs().max():.6f}")
# Output: ~0.0 (machine precision)
```

C Extended Benchmark Tables

TODO: Include forward/inversion timing (CPU+GPU), memory, coefficient counts at all group sizes. Wall-clock throughput figures.

D PCam: Extended Results

TODO: Move here from main text: full model variant descriptions (6-item list), disk bispectrum discussion, wider channel analysis, published baselines table.

E OrganMNIST3D: Extended Results

TODO: Move here from main text: published baselines table (Table 3 from original), wider channels paragraph, rotation analysis figure.

F Experimental Details

TODO: Include architecture diagrams, full hyperparameter tables, training curves, hardware specs, seeds, exact aggregation procedures for all three experiments.

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